EXTENDING DIVIDED POWERS ALONG A RING MAP: UNIQUENESS AND THREE EXISTENCE CRITERIA

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; the footnote is a complete alternative argument, transcribed as such.

Allowed standard prerequisites (editorial).

- Definition of a divided power structure. definition-divided-powers (same chapter, definitional anchor). Conditions (1)-(5), especially (3) and (4) for the uniqueness expansion.
- Generator criterion. lemma-check-on-generators (same chapter, named). Lifts the flat-case construction to a full divided power structure on IB.
- Lazard's theorem. algebra-theorem-lazard (cross-chapter, named). Writes flat B as a directed colimit of finite free modules, organizing the well-definedness of the flat-case construction. A footnote gives the alternative via algebra-lemma-flat-eq.

Source reference. Original reference 00HK: focused-seeds/geometry/S9-dpa-gamma-extends/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 058G: focused-seeds/geometry/S9-dpa-gamma-extends/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 07GL: sources/stacks-dpa/pinned-original-source.tex, source lines 253–253. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07GP: sources/stacks-dpa/pinned-original-source.tex, source lines 368–368. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let (A, I, γ) be a divided power ring. Let $A \to B$ be a ring map. If γ extends to B then it extends uniquely. Assume (at least) one of the following conditions holds*

Date: stacks-project:dpa.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) $IB = 0$,
(2) I *is principal, or*
(3) $A \to B$ *is flat.*

Then γ *extends to* B.

Proof. Any element of IB can be written as a finite sum $\sum_{i=1}^t b_i x_i$ with $b_i \in B$ and $x_i \in I$. If γ extends to $\bar\gamma$ on IB then $\bar\gamma_n(x_i) = \gamma_n(x_i)$. Thus, conditions (3) and (4) in Definition 07GL imply that

$$\bar\gamma_n(\sum\nolimits_{i=1}^t b_i x_i) = \sum\nolimits_{n_1+\ldots+n_t=n} \prod\nolimits_{i=1}^t b_i^{n_i} \gamma_{n_i}(x_i)$$

Thus we see that $\bar\gamma$ is unique if it exists.

If $IB = 0$ then setting $\bar\gamma_n(0) = 0$ works. If $I = (x)$ then we define $\bar\gamma_n(bx) = b^n\gamma_n(x)$. This is well defined: if $b'x = bx$, i.e., $(b - b')x = 0$ then

$$\begin{aligned} b^n\gamma_n(x) - (b')^n\gamma_n(x) &= (b^n - (b')^n)\gamma_n(x) \\ &= (b^{n-1} + \ldots + (b')^{n-1})(b - b')\gamma_n(x) = 0 \end{aligned}$$

because $\gamma_n(x)$ is divisible by x (since $\gamma_n(I) \subset I$) and hence annihilated by $b - b'$. Next, we prove conditions (1) – (5) of Definition 07GL. Parts (1), (2), (3), (5) are obvious from the construction. For (4) suppose that $y, z \in IB$, say $y = bx$ and $z = cx$. Then $y + z = (b + c)x$ hence

$$\begin{aligned} \bar\gamma_n(y + z) &= (b + c)^n\gamma_n(x) \\ &= \sum \frac{n!}{i!(n-i)!} b^i c^{n-i} \gamma_n(x) \\ &= \sum b^i c^{n-i} \gamma_i(x)\gamma_{n-i}(x) \\ &= \sum \bar\gamma_i(y)\bar\gamma_{n-i}(z) \end{aligned}$$

as desired.

Assume $A \to B$ is flat. Suppose that $b_1, \ldots, b_r \in B$ and $x_1, \ldots, x_r \in I$. Then

$$\bar\gamma_n(\sum b_i x_i) = \sum b_1^{e_1} \ldots b_r^{e_r} \gamma_{e_1}(x_1) \ldots \gamma_{e_r}(x_r)$$

where the sum is over $e_1 + \ldots + e_r = n$ if $\bar\gamma_n$ exists. Next suppose that we have $c_1, \ldots, c_s \in B$ and $a_{ij} \in A$ such that $b_i = \sum a_{ij} c_j$. Setting $y_j = \sum a_{ij} x_i$ we claim that

$$\sum b_1^{e_1} \ldots b_r^{e_r} \gamma_{e_1}(x_1) \ldots \gamma_{e_r}(x_r) = \sum c_1^{d_1} \ldots c_s^{d_s} \gamma_{d_1}(y_1) \ldots \gamma_{d_s}(y_s)$$

in B where on the right hand side we are summing over $d_1 + \ldots + d_s = n$. Namely, using the axioms of a divided power structure we can expand both sides into a sum with coefficients in $\mathbf{Z}[a_{ij}]$ of terms of the form $c_1^{d_1} \ldots c_s^{d_s} \gamma_{e_1}(x_1) \ldots \gamma_{e_r}(x_r)$. To see that the coefficients agree we note that the result is true in $\mathbf{Q}[x_1, \ldots, x_r, c_1, \ldots, c_s, a_{ij}]$ with γ the unique divided power structure on $(x_1, \ldots, x_r)$. By Lazard's theorem (Algebra, Theorem 058G) we can write B as a directed colimit of finite free A-modules. In particular, if $z \in IB$ is written as $z = \sum x_i b_i$ and $z = \sum x'_{i'} b'_{i'}$, then we can find $c_1, \ldots, c_s \in B$ and $a_{ij}, a'_{i'j} \in A$ such that $b_i = \sum a_{ij} c_j$ and

$b'_{i'} = \sum a'_{i'j} c_j$ such that $y_j = \sum x_i a_{ij} = \sum x'_{i'} a'_{i'j}$ holds[1]. Hence the procedure above gives a well defined map $\bar{\gamma}_n$ on IB. By construction $\bar{\gamma}$ satisfies conditions (1), (3), and (4). Moreover, for $x \in I$ we have $\bar{\gamma}_n(x) = \gamma_n(x)$. Hence it follows from Lemma 07GP that $\bar{\gamma}$ is a divided power structure on IB. □

[1]This can also be proven without recourse to Algebra, Theorem 058G. Indeed, if $z = \sum x_i b_i$ and $z = \sum x'_{i'} b'_{i'}$, then $\sum x_i b_i - \sum x'_{i'} b'_{i'} = 0$ is a relation in the A-module B. Thus, Algebra, Lemma 00HK (applied to the x_i and $x'_{i'}$ taking the place of the f_i, and the b_i and $b'_{i'}$ taking the role of the x_i) yields the existence of the $c_1, \ldots, c_s \in B$ and $a_{ij}, a'_{i'j} \in A$ as required.